

## V. Integration Test Cases

Library Management System

Test Team (Coders & Non-Coders/Specification Experts)

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**Note:** These test cases are designed to verify the interaction and data flow between different modules of the Library Management System, as per the SRS and project test plan. They focus on end-to-end user scenarios that span multiple system components.

## V. Integration Test Cases

Table 1: Integration Test Cases

| Test Case ID                                                       | Test Title / Scenario                                                                                                               | Test Steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Scenario: Full Lifecycle - From Entity Creation to Borrowal</b> |                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| TC.INT_001                                                         | End-to-end workflow of creating all required entities for a new book, registering a new member, and the member borrowing that book. | <ul style="list-style-type: none"> <li>• <b>As Librarian:</b> Login to the system.</li> <li>• Navigate to Author Management and add a new author (e.g., "J.R.R. Tolkien").</li> <li>• Navigate to Genre Management and add a new genre (e.g., "Fantasy").</li> <li>• Navigate to Book Management and add a new book ("The Hobbit"), linking the newly created author and genre. Set 'isAvailable' to true.</li> <li>• Navigate to User Management and register a new user with 'isAdmin=false' (e.g., "Member1").</li> <li>• Logout.</li> <li>• <b>As Member1:</b> Login to the system.</li> <li>• Navigate to the book list and find "The Hobbit".</li> <li>• Initiate and confirm the borrowing of "The Hobbit".</li> </ul> | <ul style="list-style-type: none"> <li>• The author is created successfully and selectable in the book form.</li> <li>• The genre is created successfully and selectable in the book form.</li> <li>• The book is created with correct author and genre associations. The book list shows its status as available.</li> <li>• The new member account is created and appears in the user list.</li> <li>• Member1 can log in successfully and is directed to the member landing page.</li> <li>• A new borrowal record is created for Member1 and "The Hobbit".</li> <li>• The 'isAvailable' status of "The Hobbit" in the Book entity is updated to 'false'.</li> <li>• The new borrowal record appears in Member1's borrowal history.</li> </ul> |
| <b>Scenario: Data Integrity and Referential Constraints</b>        |                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| TC.INT_002                                                         | Verify data integrity when a linked entity (Author) is deleted after being associated with another entity (Book).                   | <ul style="list-style-type: none"> <li>• <b>As Librarian:</b> Login.</li> <li>• Create a new, unique Author (e.g., "IntegTest Author").</li> <li>• Create a new Book (e.g., "IntegTest Book") and link it to "IntegTest Author".</li> <li>• Verify the book details page shows "IntegTest Author" as the author.</li> <li>• Navigate to Author Management and delete "IntegTest Author".</li> <li>• Navigate back to Book Management and view the details of "IntegTest Book".</li> </ul>                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>• The Author is successfully deleted.</li> <li>• The Book "IntegTest Book" still exists in the system.</li> <li>• The 'authorId' field for "IntegTest Book" is handled gracefully (e.g., set to null or an empty value), as the data model indicates it's optional. The book details page no longer displays the deleted author's name.</li> <li>• The system does not crash and remains in a consistent state.</li> </ul>                                                                                                                                                                                                                                                                                 |
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| Test Case ID                                                  | Test Title / Scenario                                                                                 | Test Steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TC_INT_003                                                    | Verify that deleting a User with active Borrowals does not violate data consistency.                  | <ul style="list-style-type: none"> <li>• <b>As Librarian:</b> Login.</li> <li>• Ensure a specific Member (e.g., "MemberToDelete") has at least one active borrowal record. If not, create one.</li> <li>• Navigate to User Management.</li> <li>• Attempt to delete the user "MemberToDelete".</li> </ul>                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>• <b>Expected (Option A - Deletion Blocked):</b> The system prevents the deletion and displays an error message indicating the user has active borrowals that must be resolved first.</li> <li>• <b>Expected (Option B - Soft Delete/Orphaning):</b> The user is deleted. The 'memberId' in the Borrowal record remains but now points to a non-existent user. The borrowal record is still visible to the Librarian, but may indicate the user is deleted.</li> <li>• (The exact expected result depends on the implemented business rule for referential integrity, which should be consistent).</li> </ul> |
| <b>Scenario: State Transitions Across Modules</b>             |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| TC_INT_004                                                    | Verify that the state of a Book (isAvailable) is correctly updated by actions in the Borrowal module. | <ul style="list-style-type: none"> <li>• <b>Prerequisite:</b> A book "StateTest Book" exists and is available. A "StateTest Member" exists.</li> <li>• <b>As Member:</b> Login as "StateTest Member". Borrow "StateTest Book".</li> <li>• <b>As Librarian:</b> Login. Navigate to Book Management. View the status of "StateTest Book".</li> <li>• Try to create a new borrowal for "StateTest Book" for another member.</li> <li>• Navigate to Borrowal Management. Find the borrowal record for "StateTest Book" and update its status to "Returned".</li> <li>• Navigate back to Book Management and view the status of "StateTest Book" again.</li> </ul> | <ul style="list-style-type: none"> <li>• After the member borrows the book, the book's 'isAvailable' status immediately changes to 'false'.</li> <li>• The Librarian cannot create a new borrowal for the now-unavailable book. The UI should prevent this action.</li> <li>• After the Librarian marks the borrowal as "Returned", the borrowal record's status field is updated.</li> <li>• The 'isAvailable' status of "StateTest Book" in the Book module reverts to 'true'.</li> <li>• The data remains consistent between the Book and Borrowal modules throughout the workflow.</li> </ul>                                                    |
| <b>Scenario: Role-Based Access Control (RBAC) Integration</b> |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| Test Case ID | Test Title / Scenario                                                                                                          | Test Steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| TC_INT_005   | Verify that user roles created in the User module correctly restrict access to features in other modules (e.g., Book, Author). | <ul style="list-style-type: none"> <li>• <b>As Librarian:</b> Login. Create a new user "TestMember" with the role of Member ('isAdmin=false').</li> <li>• Logout.</li> <li>• <b>As Member:</b> Login as "TestMember".</li> <li>• Attempt to access the User Management page via direct URL navigation (e.g., '/users').</li> <li>• Navigate to the Books list.</li> <li>• Check for the presence of "Add New Book", "Edit", or "Delete" buttons/options.</li> <li>• Using an API client (e.g., Postman), attempt to send a 'POST' request to '/api/book/add' using the session/token of "TestMember".</li> </ul> | <ul style="list-style-type: none"> <li>• The Member login is successful.</li> <li>• Direct navigation to the '/users' URL is blocked, and the user is redirected to the member dashboard or an error page.</li> <li>• The UI for the Books list does not show any administrative controls (Add, Edit, Delete).</li> <li>• The API request to 'POST /api/book/add' fails with an authorization error (e.g., 403 Forbidden or 401 Unauthorized), demonstrating that the access control middleware correctly uses the user's role from the session.</li> </ul> |